0.1 The test cases of Lynch and Gray

Lynch and Gray [2] derive a number of analytic solutions to the two-dimensional, depth-averaged, linear shallow water equations (SWE) that provide a systematic and rigorous means of verifying numerical models for the SWE. The problems considered incorporate several features that make them challenging test cases, including time periodic open boundary forcing, bottom friction, wind stress, spatially varying bathymetry, and both polar and Cartesian geometry domains.

To begin, the domains of the Cartesian and polar geometry test cases are shown in Figure 1. No-normal flow conditions (i.e., solid wall or land boundaries) are specified along the boundaries of the domain at $x = x_1$, $y = 0$ and $y = L$ ($r = r_1$, $\theta = 0$ and $\theta = \phi$, respectively, for the polar geometry case). A surface elevation is specified along the "open ocean" boundary along $x = x_2$ ($r = r_2$) described by a periodic tidal forcing function of the general form

$$\zeta(x_2, y, t) = \mathrm{Re}\left\{\zeta_0\,(y)\, e^{i\omega t}\right\}, \tag{1}$$

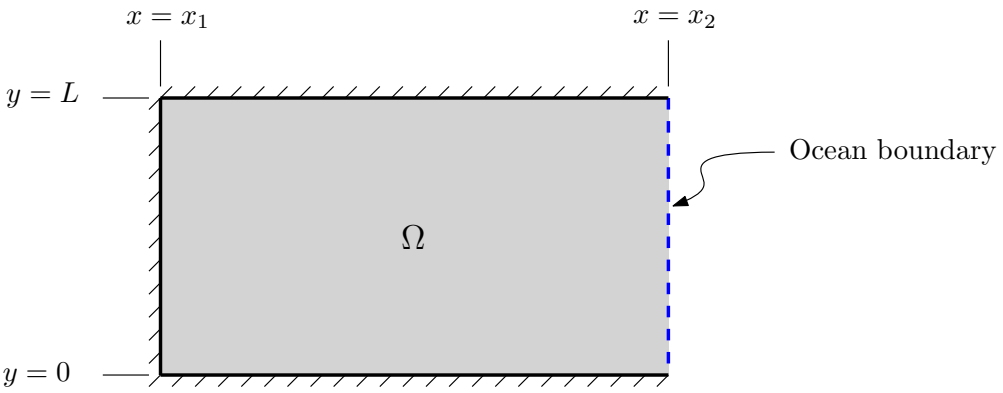
where ζ_0 is, in general, a complex function representing the tidal forcing amplitude and phase, ω is the (angular) frequency of the tidal forcing function and $i = \sqrt{-1}$ (for the polar geometry case, x_2 and y are replaced by r_2 and θ, respectively, in (1)).

To complete the description of the problem domains, the bathymetry along constant values of y or θ is given by the following power law

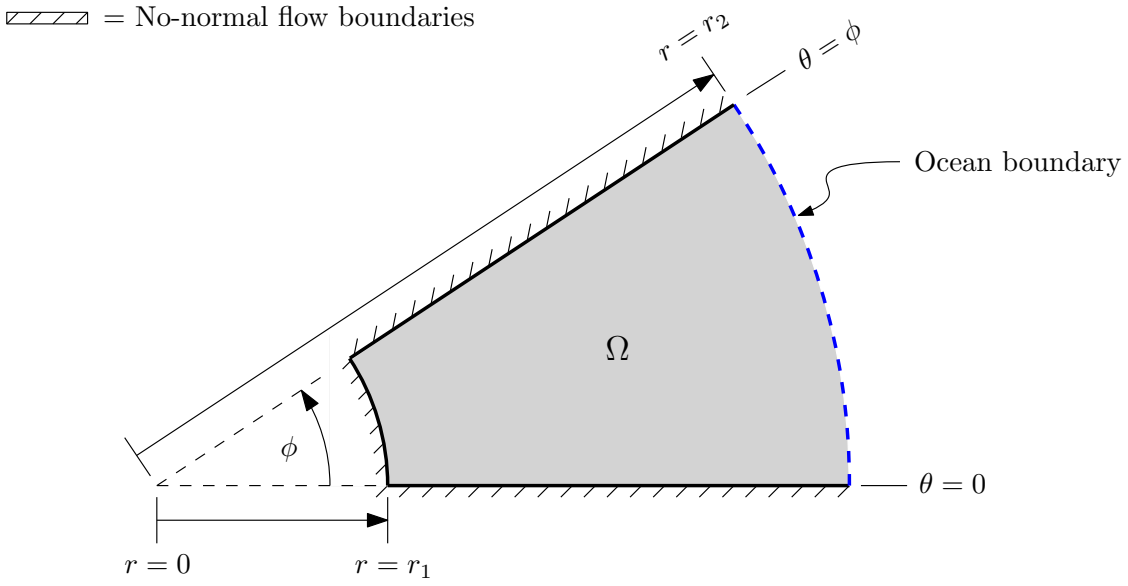
$$h(x) = h_0 x^n, \tag{2}$$

where h_0 is a specified real positive constant and n, also specified, is a real constant that is not necessarily an integer (again, x is replaced by r for the polar case). In Figure 2, cross sections of the bathymetry along constant values of y and θ are shown for the cases $n = 0, 1, \pm 2$.

In general, the solutions for these test cases involve the evaluation of an infinite series. However, in the case that Equation (1) is independent of y or θ, the analytic solutions are greatly simplified — all the terms except the first in the infinite series being zero. For this special case, Lynch and Gray give complete solutions for flat, linear, and quadratic bathymetry, i.e., $n = 0, 1, \pm 2$ in Equation (2). As the original publication of Lynch and Gray contains errors for the reverse quadratic bathymetry cases ($n = -2$), and for the sake of completeness, we present the correct solutions for all the test cases in both Cartesian and polar geometries in Tables 1 and 2, respectively, with the correct solutions for the $n = -2$ cases being obtained from [1]. (We also correct a minor typo in the solution of the surface elevation for the $n = 0$ polar geometry test case). We remark that the solutions that involve Bessel functions of order p of the first (Jp) and second (Yp) kind can easily be evaluated numerically using, for example, Matlab's built-in Bessel functions `besselj` and `bessely`, respectively. Finally, we note that Lynch and Gray also derive solutions with the addition of a unit wind stress, which we have not considered here.



Cartesian geometry domain



Polar geometry domain

Figure 1: Plan views of the Cartesian and polar geometry test case domains of Lynch and Gray.

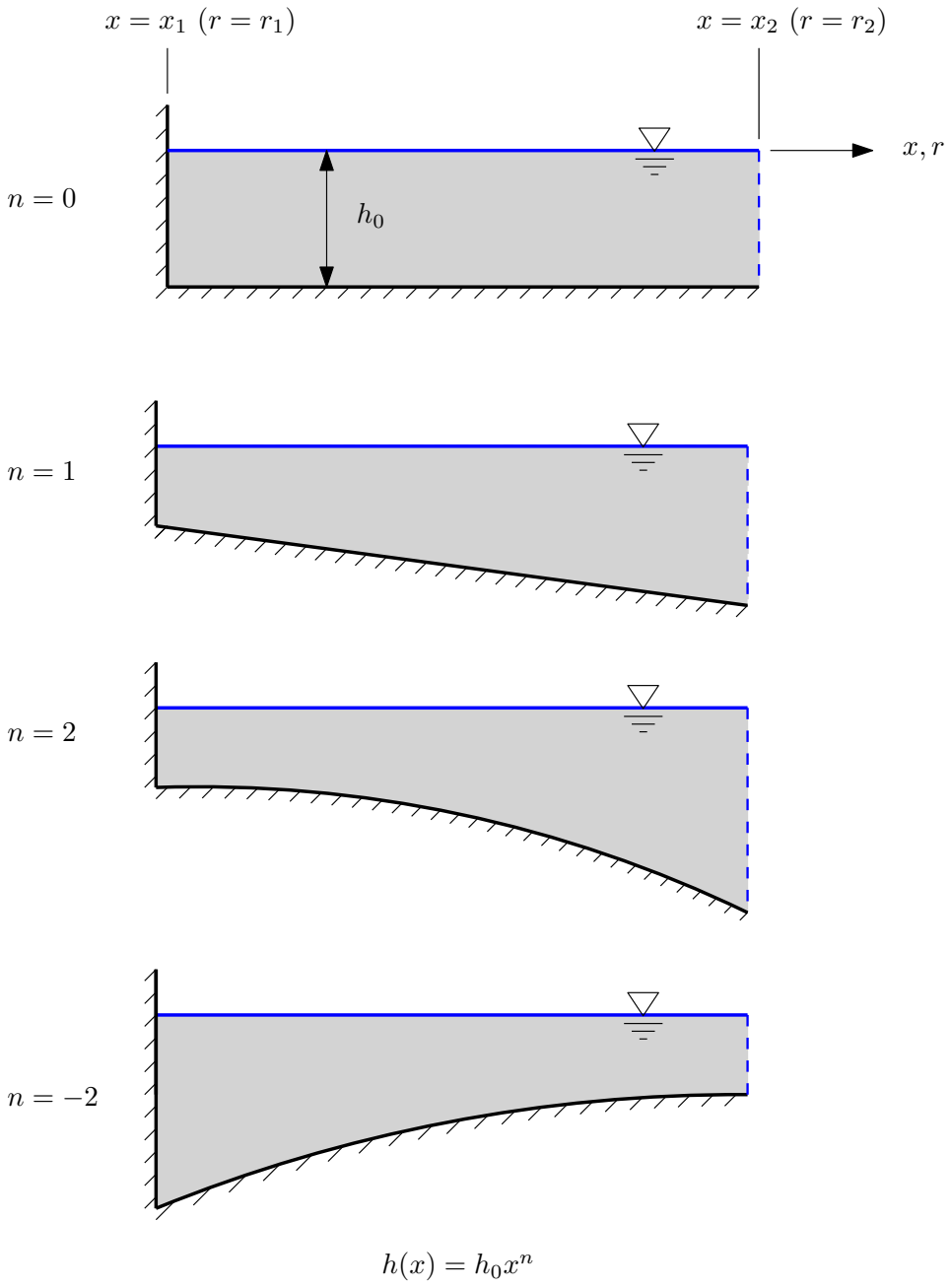


Figure 2: Cross-sectional views of the bathymetry along constant values of y and θ for the test cases of Lynch and Gray for $n = 0, 1, \pm 2$.

Bathymetry	Solutions	Constants
$n=0$	$\zeta(x,t)=\mathrm{Re}\left\{\zeta_0\left[\dfrac{\cos(\beta(x-x_1))}{\cos(\beta(x_2-x_1))}\right]e^{i\omega t}\right\}$ $u(x,t)=\mathrm{Re}\left\{-\zeta_0\left[\dfrac{\sin(\beta(x-x_1))}{\cos(\beta(x_2-x_1))}\right]\dfrac{i\omega}{\beta h_0}e^{i\omega t}\right\}$	
$n=1$	$\zeta(x,t)=\mathrm{Re}\left\{\left[AJ_0\left(2\beta\sqrt{x}\right)+BY_0\left(2\beta\sqrt{x}\right)\right]e^{i\omega t}\right\}$ $u(x,t)=\mathrm{Re}\left\{-\dfrac{1}{\sqrt{x}}\left[AJ_1\left(2\beta\sqrt{x}\right)+BY_1\left(2\beta\sqrt{x}\right)\right]\dfrac{i\omega}{\beta h_0}e^{i\omega t}\right\}$	$A=\dfrac{\zeta_0 Y_1\left(2\beta\sqrt{x_1}\right)}{D},\quad B=\dfrac{\zeta_0 J_1\left(2\beta\sqrt{x_1}\right)}{D}$ $D=J_0\left(2\beta\sqrt{x_2}\right)Y_1\left(2\beta\sqrt{x_1}\right)-Y_0\left(2\beta\sqrt{x_2}\right)J_1\left(2\beta\sqrt{x_1}\right)$
$n=2$	$\zeta(x,t)=\mathrm{Re}\left\{\left[Ax^{s_1}+Bx^{s_2}\right]e^{i\omega t}\right\}$ $u(x,t)=\mathrm{Re}\left\{\left[As_1x^{s_1-1}+Bs_2x^{s_2-1}\right]\dfrac{i\omega}{\beta^2 h_0}e^{i\omega t}\right\}$	$A=\dfrac{\zeta_0 s_2 x_1^{s_2}}{D},\quad B=\dfrac{-\zeta_0 s_1 x_1^{s_1}}{D}$ $D=s_2x_2^{s_1}x_1^{s_2}-s_1x_1^{s_1}x_2^{s_2},\quad s_1,s_2=-1/2\pm\sqrt{1/4-\beta^2}$
$n=-2$	$\zeta(x,t)=\mathrm{Re}\left\{x^{3/2}\left[AJ_{3/4}\left(\frac{\beta x^2}{2}\right)+BJ_{-3/4}\left(\frac{\beta x^2}{2}\right)\right]e^{i\omega t}\right\}$ $u(x,t)=\mathrm{Re}\left\{x^{5/2}\left[AJ_{-1/4}\left(\frac{\beta x^2}{2}\right)+BJ_{1/4}\left(\frac{\beta x^2}{2}\right)\right]\dfrac{i\omega}{\beta h_0}e^{i\omega t}\right\}$	$A=\dfrac{\zeta_0 J_{1/4}(\beta x_1^2/2)}{D},\quad B=\dfrac{\zeta_0 J_{-1/4}(\beta x_1^2/2)}{D}$ $D=x_2^{3/2}\left[J_{1/4}\left(\frac{\beta x_1^2}{2}\right)J_{3/4}\left(\frac{\beta x_2^2}{2}\right)+J_{-3/4}\left(\frac{\beta x_2^2}{2}\right)J_{-1/4}\left(\frac{\beta x_1^2}{2}\right)\right]$

Notes: $\beta^2=\left(\omega^2-i\omega\tau\right)/gh_0$, $J_p=$ Bessel function of the first kind, $Y_p=$ Bessel function of the second kind

Table 1: The Cartesian geometry test cases of Lynch and Gray [2]

Bathymetry	Solutions	Constants
$n=0$	$\zeta(r,t)=\mathrm{Re}\left\{\left[AJ_0(\beta r)+BY_0(\beta r)\right]e^{i\omega t}\right\}$ $v_r(r,t)=\mathrm{Re}\left\{-\left[AJ_1(\beta r)+BY_1(\beta r)\right]\frac{i\omega}{\beta h_0}e^{i\omega t}\right\}$	$A=\frac{\zeta_0 Y_1(\beta r_1)}{D},\quad B=\frac{-\zeta_0 J_1(\beta r_1)}{D}$ $D=J_0(\beta r_2)Y_1(\beta r_1)-J_1(\beta r_1)Y_0(\beta r_2)$
$n=1$	$\zeta(r,t)=\mathrm{Re}\left\{\frac{1}{\sqrt{r}}\left[AJ_1(2\beta\sqrt{r})+BY_1(2\beta\sqrt{r})\right]e^{i\omega t}\right\}$ $v_r(r,t)=\mathrm{Re}\left\{\frac{1}{r}\left[-AJ_2(2\beta\sqrt{r})-BY_2(2\beta\sqrt{r})\right]\frac{i\omega}{\beta h_0}e^{i\omega t}\right\}$	$A=\frac{\zeta_0\sqrt{r_2}Y_2(2\beta\sqrt{r_1})}{D},\quad B=\frac{-\zeta_0\sqrt{r_2}J_2(2\beta\sqrt{r_1})}{D}$ $D=J_1(2\beta\sqrt{r_2})Y_2(2\beta\sqrt{r_1})-J_2(2\beta\sqrt{r_1})Y_1(2\beta\sqrt{r_2})$
$n=1$	$\zeta(r,t)=\mathrm{Re}\left\{\left[Ar^{s_1}+Br^{s_2}\right]e^{i\omega t}\right\}$ $v_r(r,t)=\mathrm{Re}\left\{\left[As_1r^{s_1-1}+Bs_2r^{s_2-1}\right]\frac{i\omega}{\beta^2 h_0}e^{i\omega t}\right\}$	$A=\frac{\zeta_0 s_2 r_1^{s_2}}{D},\quad B=\frac{-\zeta_0 s_1 r_1^{s_1}}{D}$ $D=s_2r_2^{s_1}r_1^{s_2}-s_1r_1^{s_1}r_2^{s_2},\quad s_1,s_2=-1\pm\sqrt{1-\beta^2}$
$n=-2$	$\zeta(x,t)=\mathrm{Re}\left\{\zeta_0\left[\frac{\cos\left(\frac{\beta}{2}(r^2-r_1^2)\right)}{\cos\left(\frac{\beta}{2}(r_2^2-r_1^2)\right)}\right]e^{i\omega t}\right\}$ $v_r(x,t)=\mathrm{Re}\left\{-\zeta_0 r\left[\frac{\sin\left(\frac{\beta}{2}(r^2-r_1^2)\right)}{\cos\left(\frac{\beta}{2}(r_2^2-r_1^2)\right)}\right]\frac{i\omega}{\beta h_0}e^{i\omega t}\right\}$	

Notes: $\beta^2=\left(\omega^2-i\omega\tau\right)/gh_0$, $J_p=$ Bessel function of the first kind, $Y_p=$ Bessel function of the second kind, $v_r=$ radial velocity

Table 2: The polar geometry test cases of Lynch and Gray [2]

References

References

[1] C. L. Chen, *Analytic solutions for tidal model testing*, Journal of hydraulic engineering, 115 (1989), pp. 1707–1714.

[2] D. R. Lynch and W. G. Gray, *Analytic solutions for computer flow model testing*, Journal of the hydraulics division, 104 (1978), pp. 1409–1428.